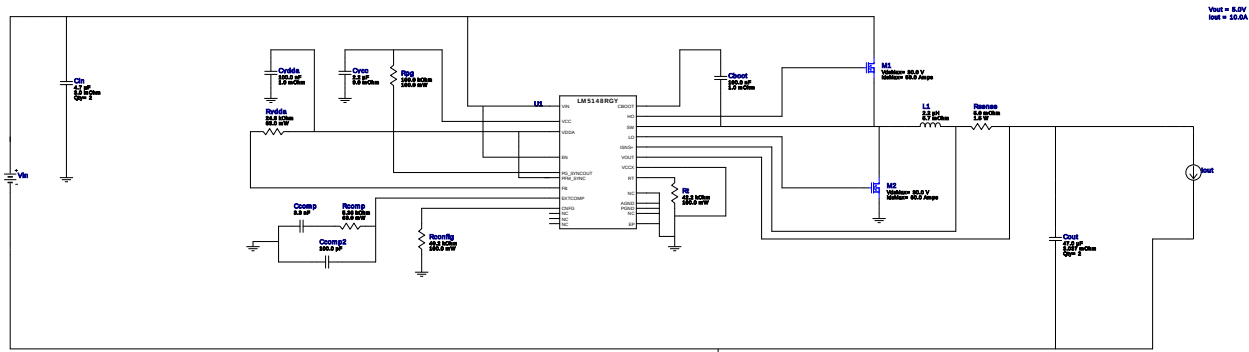


WEBENCH® Design Report

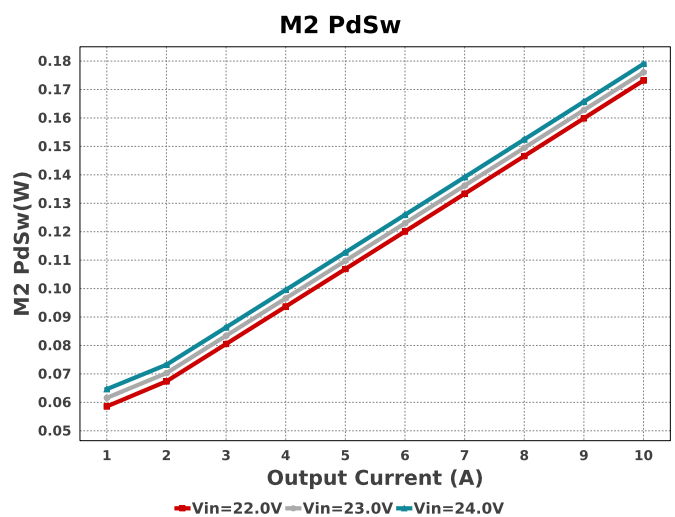
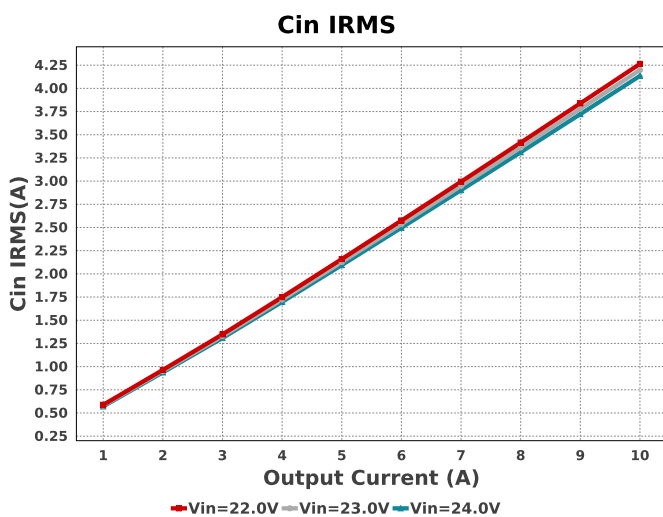
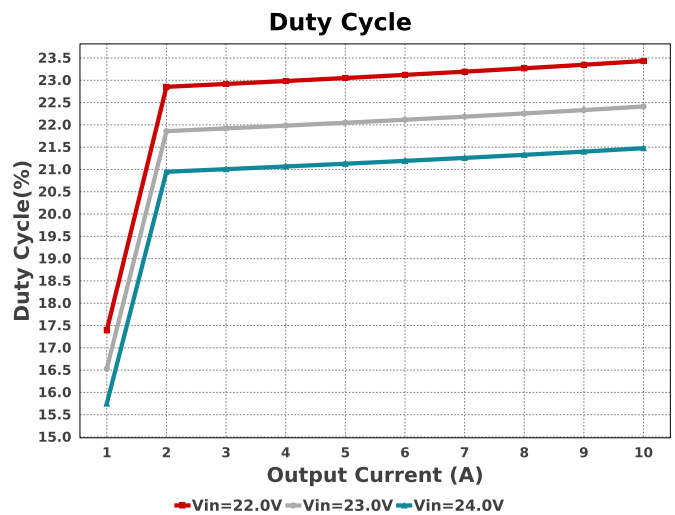
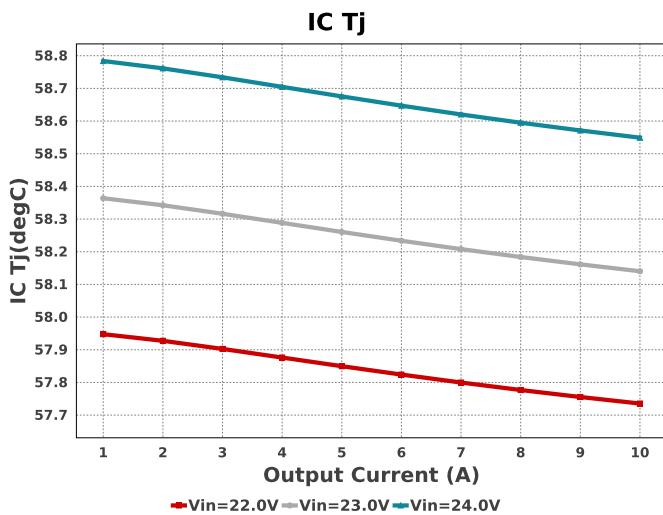
Design : 18 LM5148RGYR
LM5148RGYR 22V-24V to 5.00V @ 10A

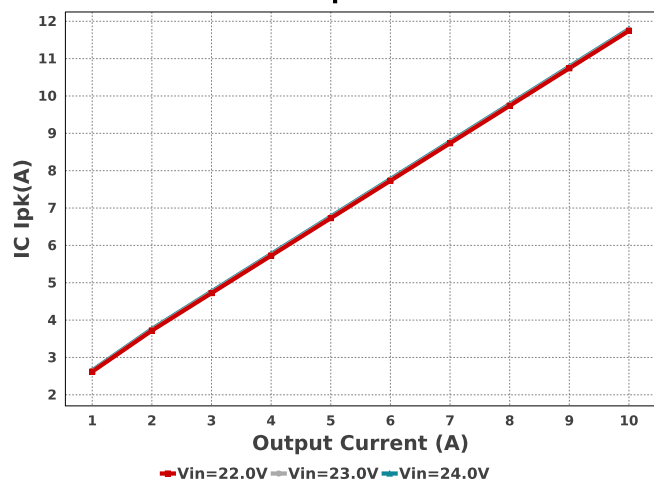


Electrical BOM

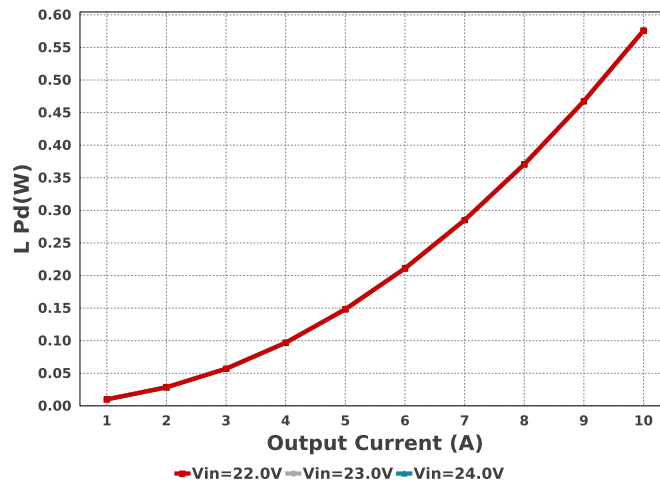
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	MuRata	GRM155R71A104KA01D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	\$0.01	0402 3 mm ²
Ccomp	TDK	CGA4C2C0G1H332J060AA Series= C0G/NP0	Cap= 3.3 nF VDC= 50.0 V IRMS= 0.0 A	1	\$0.04	0805 7 mm ²
Ccomp2	Samsung Electro-Mechanics	CL21C101JBANNNC Series= C0G/NP0	Cap= 100.0 pF VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	0805 7 mm ²
Cin	MuRata	GRM31CR71H475KA12L Series= X7R	Cap= 4.7 uF ESR= 3.0 mOhm VDC= 50.0 V IRMS= 4.98 A	2	\$0.10	1206 11 mm ²
Cout	MuRata	GRM32ER61C476KE15L Series= X5R	Cap= 47.0 uF ESR= 3.037 mOhm VDC= 16.0 V IRMS= 4.59346 A	2	\$0.17	1210_280 15 mm ²
Cvcc	MuRata	GRM188R71A225KE15D Series= X7R	Cap= 2.2 uF ESR= 9.0 mOhm VDC= 10.0 V IRMS= 3.3 A	1	\$0.02	0603 5 mm ²
Cvdda	MuRata	GRM155R71A104KA01D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	\$0.01	0402 3 mm ²
L1	Coilcraft	XAL7070-222MEB	L= 2.2 uH 5.7 mOhm	1	\$1.19	XAL7070 87 mm ²
M1	Texas Instruments	CSD17578Q5A	VdsMax= 30.0 V IdsMax= 59.0 Amps	1	\$0.15	TRANS_NexFET_Q5A 55 mm ²
M2	Texas Instruments	CSD17577Q5A	VdsMax= 30.0 V IdsMax= 60.0 Amps	1	\$0.17	TRANS_NexFET_Q5A 55 mm ²

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Rcomp	Vishay-Dale	CRCW04025K36FKED Series= CRCW..e3	Res= 5.36 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rconfig	Yageo	RC0603FR-0740K2L Series= ?	Res= 40.2 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Rpg	Vishay-Dale	CRCW0603100KFKEA Series= CRCW..e3	Res= 100.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Rsense	Stackpole Electronics Inc	CSNL2010FT5L00 Series= ?	Res= 5.0 mOhm Power= 1.5 W Tolerance= 1.0%	1	\$0.19	2010 32 mm ²
Rt	Vishay-Dale	CRCW060342K2FKEA Series= CRCW..e3	Res= 42.2 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Rvdda	Vishay-Dale	CRCW040224K3FKED Series= CRCW..e3	Res= 24.3 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
U1	Texas Instruments	LM5148RGYR	Switcher	1	\$0.76	RGY0024F 43 mm ²

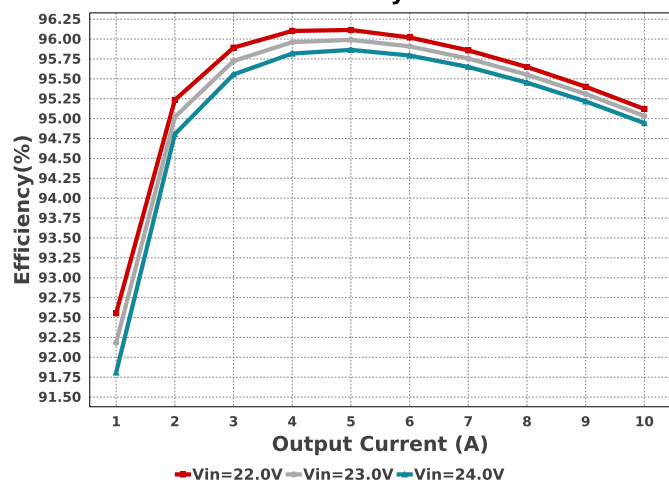


IC I_{pk}

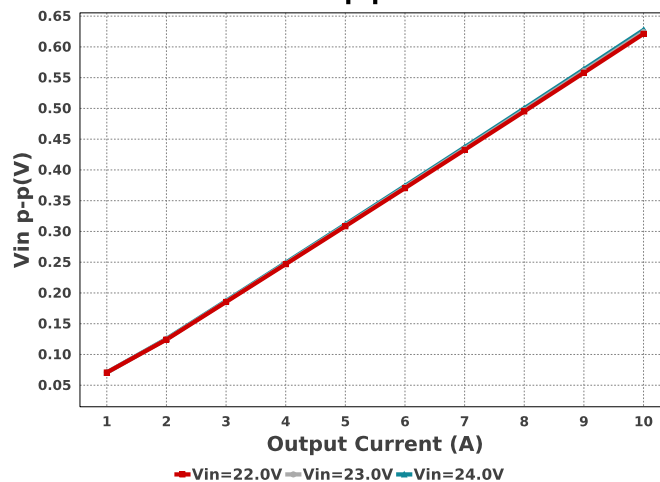
L Pd



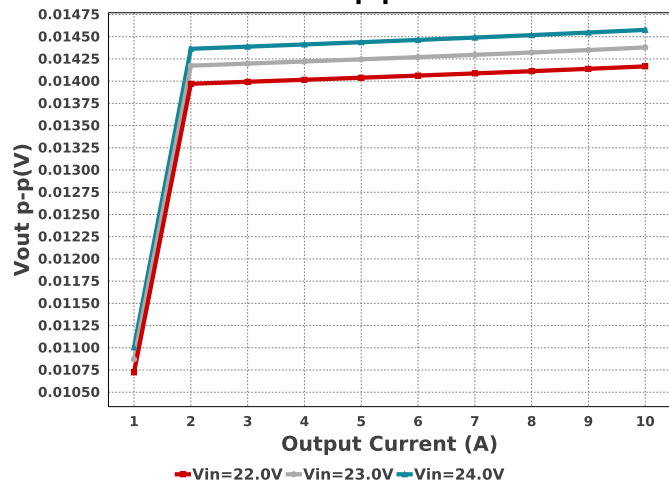
Efficiency



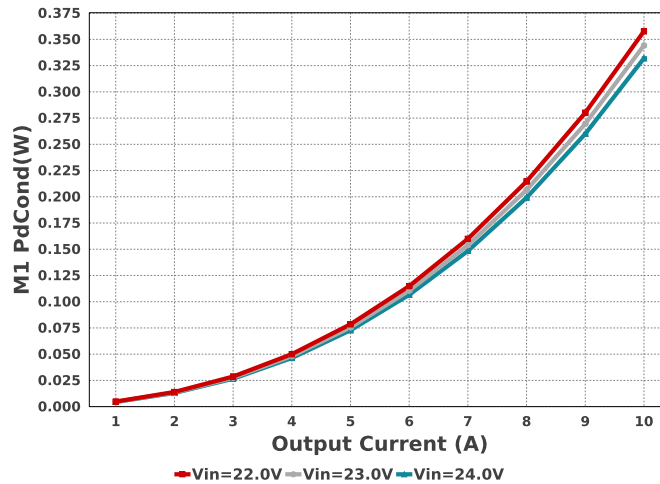
Vin p-p

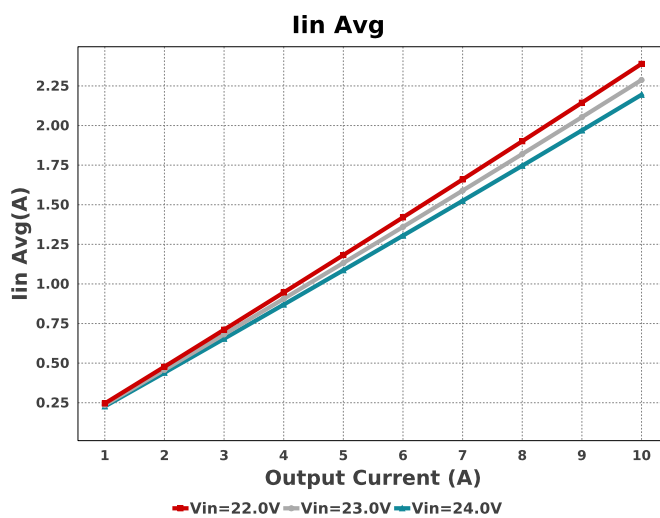
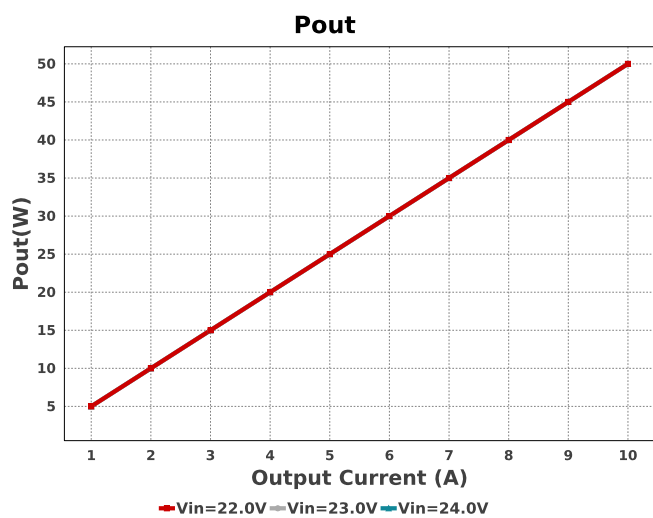
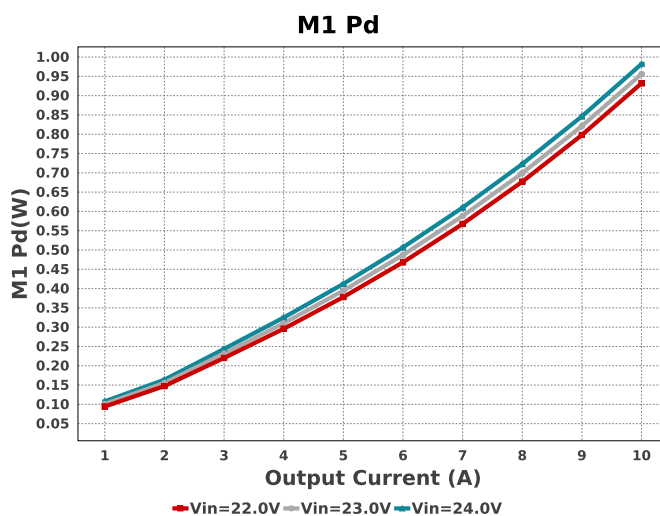
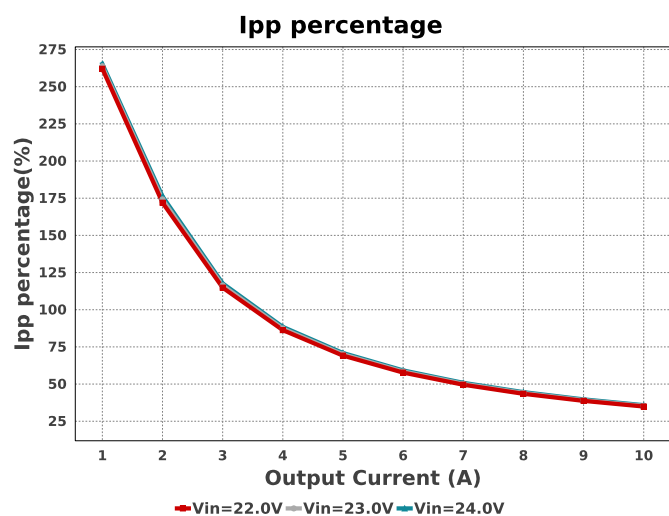
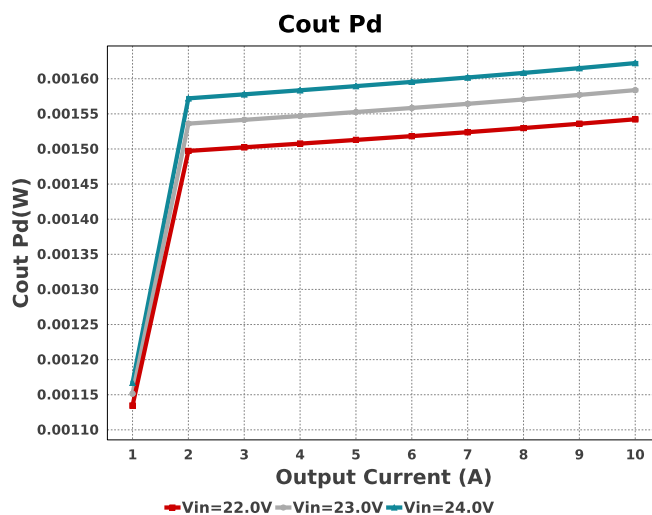
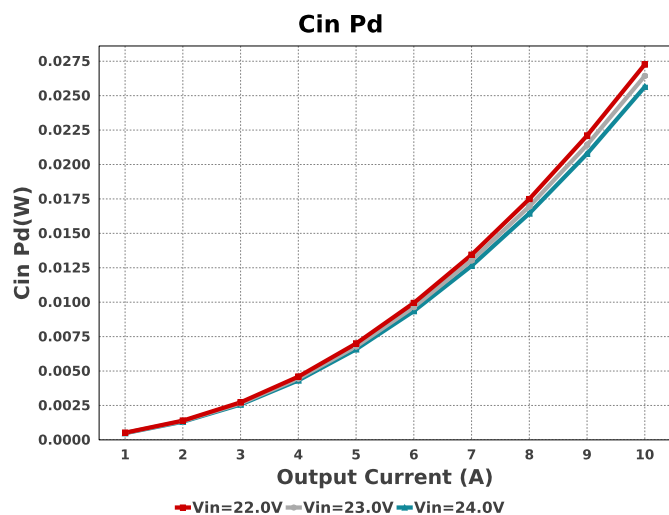


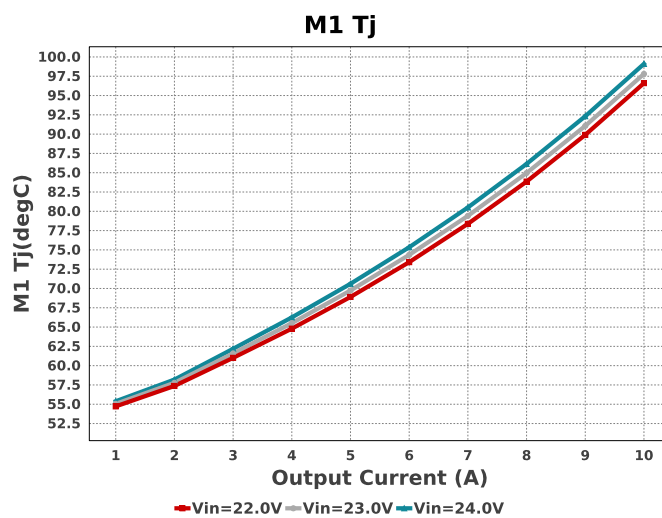
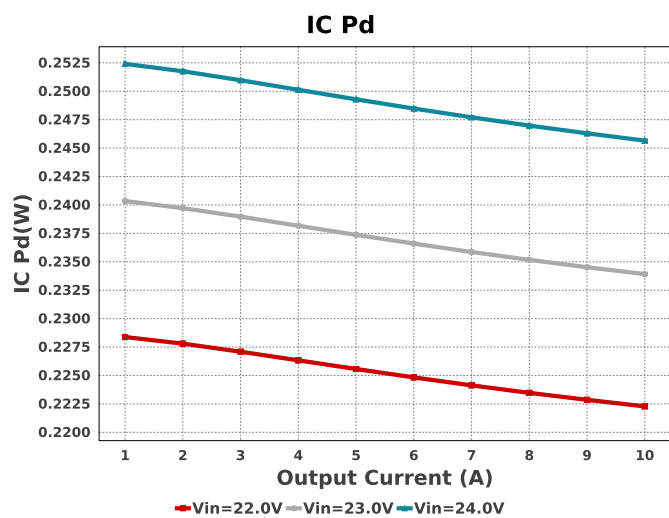
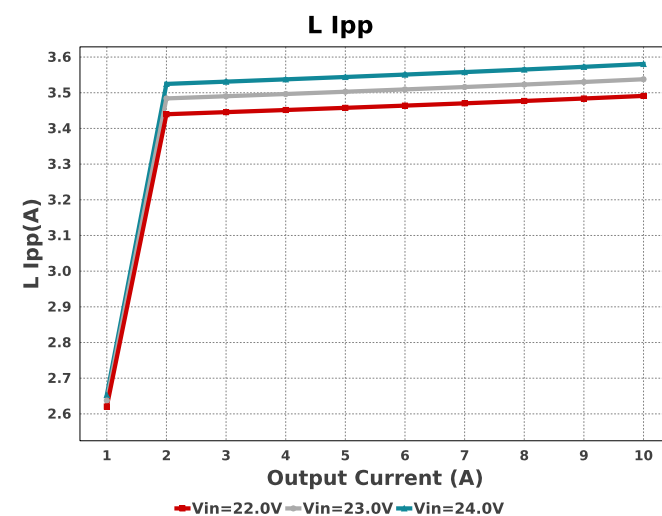
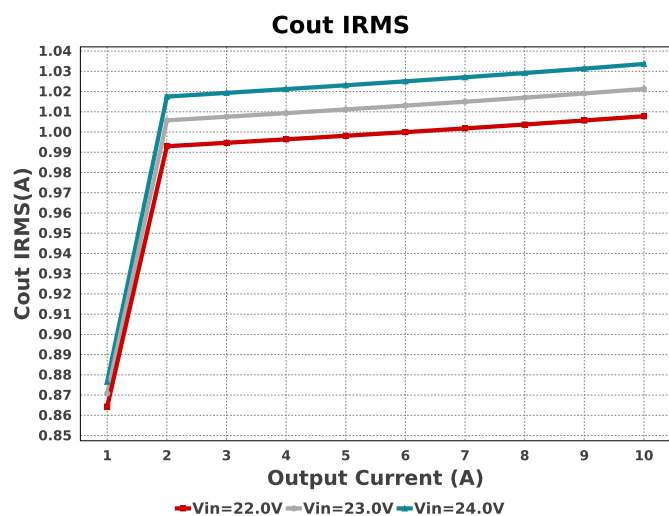
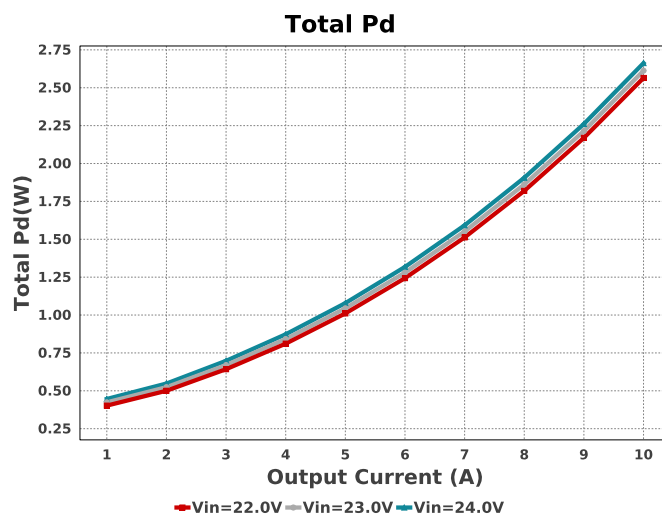
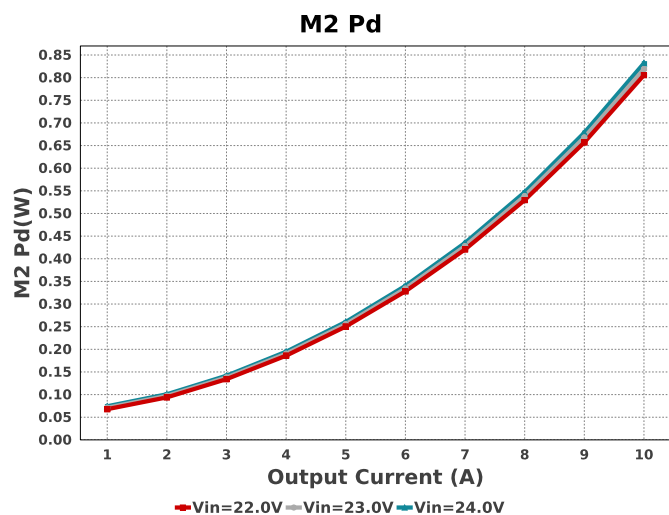
Vout p-p

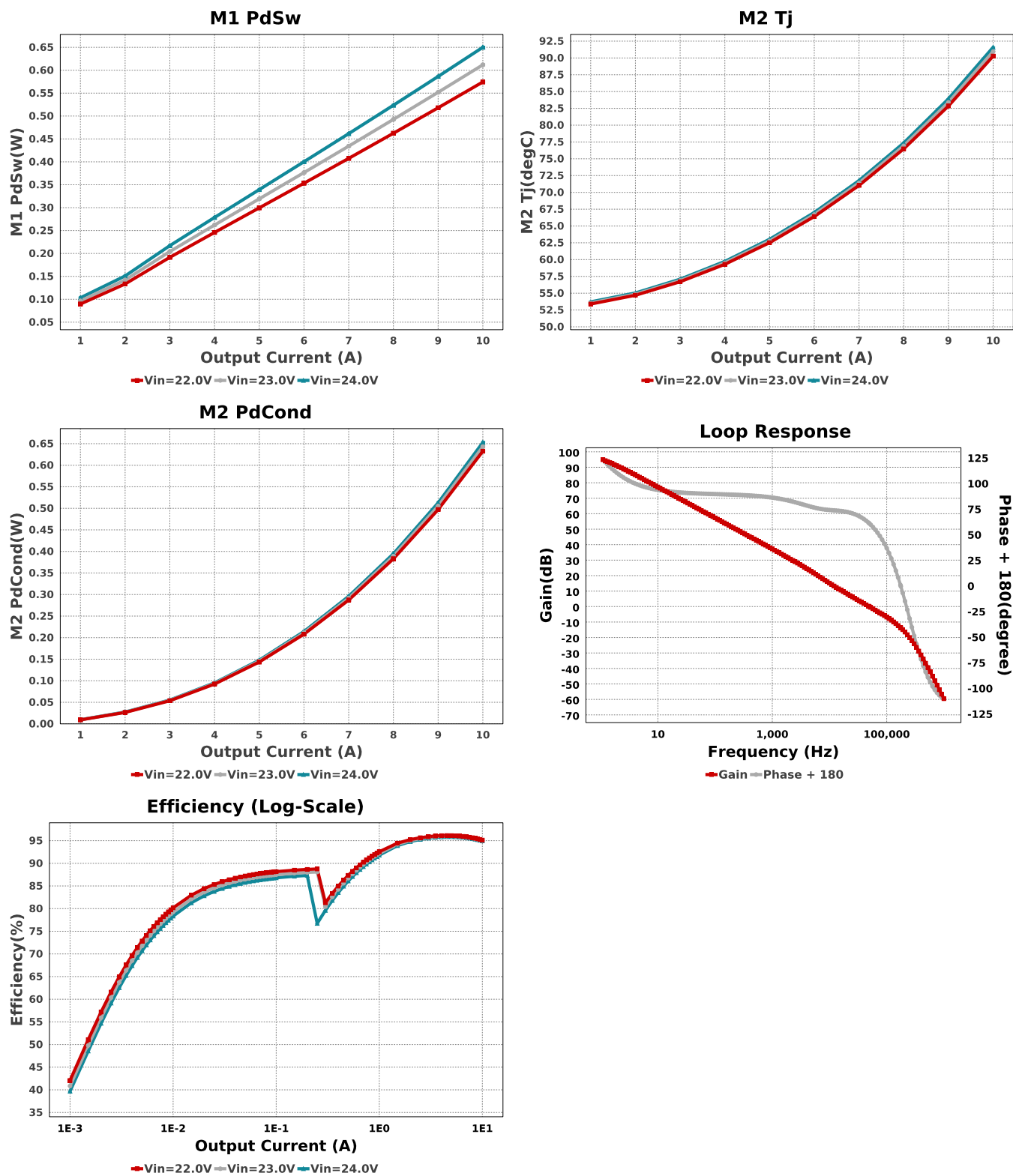


M1 PdCond









Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	4.134 A	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	25.641 mW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	1.034 A	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	1.622 mW	Capacitor	Output capacitor power dissipation
5.	IC Ipk	11.79 A	IC	Peak switch current in IC
6.	IC Pd	245.66 mW	IC	IC power dissipation
7.	IC Tj	58.549 degC	IC	IC junction temperature
8.	IC Tolerance	50.0 mV	IC	IC Feedback Tolerance
9.	ICThetaJA Effective	34.8 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
10.	Iin Avg	2.194 A	IC	Average input current

#	Name	Value	Category	Description
11.	Ipp percentage	35.806 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
12.	L Ipp	3.581 A	Inductor	Peak-to-peak inductor ripple current
13.	L Pd	576.09 mW	Inductor	Inductor power dissipation
14.	M1 Pd	982.06 mW	Mosfet	M1 MOSFET total power dissipation
15.	M1 PdCond	331.98 mW	Mosfet	M1 MOSFET conduction losses
16.	M1 PdSw	650.09 mW	Mosfet	M1 MOSFET switching losses
17.	M1 Tj	99.103 degC	Mosfet	M1 MOSFET junction temperature
18.	M2 Pd	831.75 mW	Mosfet	M2 MOSFET total power dissipation
19.	M2 PdCond	652.74 mW	Mosfet	M2 MOSFET conduction losses
20.	M2 PdSw	179.01 mW	Mosfet	M2 MOSFET switching losses
21.	M2 Tj	91.588 degC	Mosfet	M2 MOSFET junction temperature
22.	Cin Pd	25.641 mW	Power	Input capacitor power dissipation
23.	Cout Pd	1.622 mW	Power	Output capacitor power dissipation
24.	IC Pd	245.66 mW	Power	IC power dissipation
25.	L Pd	576.09 mW	Power	Inductor power dissipation
26.	M1 Pd	982.06 mW	Power	M1 MOSFET total power dissipation
27.	M1 PdCond	331.98 mW	Power	M1 MOSFET conduction losses
28.	M1 PdSw	650.09 mW	Power	M1 MOSFET switching losses
29.	M2 Pd	831.75 mW	Power	M2 MOSFET total power dissipation
30.	M2 PdCond	652.74 mW	Power	M2 MOSFET conduction losses
31.	M2 PdSw	179.01 mW	Power	M2 MOSFET switching losses
32.	Total Pd	2.663 W	Power	Total Power Dissipation
33.	BOM Count	19	System	Total Design BOM count
34.	Cross Freq	49.472 kHz	Information System	Bode plot crossover frequency
35.	Duty Cycle	21.477 %	Information System	Duty cycle
36.	Efficiency	94.944 %	Information System	Steady state efficiency
37.	FootPrint	367.0 mm ²	Information System	Total Foot Print Area of BOM components
38.	Frequency	512.295 kHz	Information System	Switching frequency
39.	Gain Marg	-13.777 dB	Information System	Bode Plot Gain Margin
40.	Iout	10.0 A	Information System	Iout operating point
41.	Iout transient step used for Cout calculations	5.0 A	Information System	Custom Transient current step requirement that was used for Cout selection (A).
42.	Low Freq Gain	94.908 dB	Information System	Gain at 1Hz
43.	Mode	CCM	Information System	Conduction Mode
44.	Overshoot Value	86.761 mV	Information System	Theoretical Vout Overshoot Value
45.	Phase Marg	61.54 deg	Information System	Bode Plot Phase Margin
46.	Pout	50.0 W	Information System	Total output power
47.	Total BOM	\$3.14	Information System	Total BOM Cost
48.	Undershoot Value	144.718 mV	Information System	Theoretical Vout Undershoot Value
49.	Vin	24.0 V	Information System	Vin operating point
50.	Vin p-p	627.486 mV	Information System	Peak-to-peak input voltage
51.	Vout	5.0 V	Information System	Operational Output Voltage
52.	Vout Ripple requirement used for Cout calculations	1.0 %	Information System	Custom maximum output ripple requirement that was used for Cout selection(% of Vout).
53.	Vout Tolerance	1.0 %	Information System	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
54.	Vout p-p	14.577 mV	Information System	Peak-to-peak output ripple voltage
55.	Vout transient requirement used for Cout calculations	3.0 %	Information System	Custom Transient voltage change requirement that was used for Cout selection (% of Vout).

Design Inputs

Name	Value	Description
Iout	10.0	Maximum Output Current

Name	Value	Description
VinMax	24.0	Maximum input voltage
VinMin	22.0	Minimum input voltage
Vout	5.0	Output Voltage
base_pn	LM5148	Base Product Number
source	DC	Input Source Type
Ta	50.0	Ambient temperature
UserFsw	500.0 k	Customer Selected Frequency

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

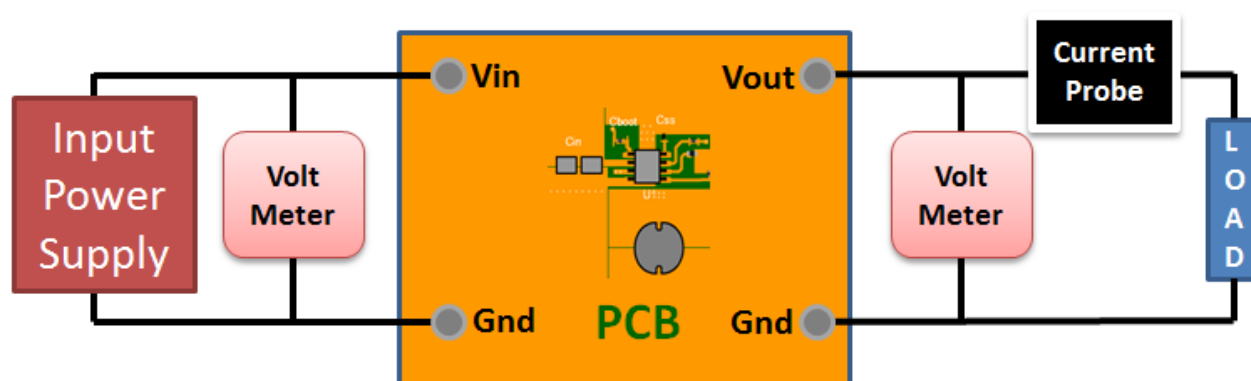
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 22.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 14CACC71A79F24D6AF9667573EFEAE90[v1]
2. **LM5148** Product Folder : <http://www.ti.com/product/LM5148> : contains the data sheet and other resources.

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